Payment and Settlement Facility-Class Design Basis

The First Facility Class of the Financial Services Sector Design Basis

DBA-FIN-PS-FC1 — Facility-Class Design Basis, DBA-FIN Series

Issued Under: Financial Services Sector Framework (DBA-FIN_Sector_Framework v0.4); DBA-MA Parent Framework

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This version is a member of CIKR Design Basis Examples, Edition 1 (November 2026) (https://doi.org/10.5281/zenodo.23863986), a dated edition that freezes, for each of fifteen critical infrastructure sectors, a sector framework, a facility class and a reference event as they stood on 1 November 2026. The edition is not falsified by a later revision of this document, whose working master continues to move. It is a working draft, marked DRAFT on its cover.

1. Purpose and Scope

1.1 Purpose

This document sets out the design basis for payment and settlement, the first facility class in the DBA-FIN series. The *Financial Services Sector Framework* (DBA-FIN_Sector_Framework) states the sector's design basis in general terms, and this document applies it to one function: the movement of value between institutions with finality. It names the events a payment system and the institutions that use it must withstand, traces how each event reaches the counterparties, the markets and the public who depend on settlement completing, sets the parameters that size the class's protection, and states what the system and its participants must each be able to do when settlement cannot complete, or completes on an instruction that was false. It describes no particular system. A design basis for a specific payment system, operator or participant starts from this document and adds only what is particular to it.

The framework calls Financial Services the first consumer sector in the corpus: it produces no physical thing, and its facility classes are financial functions running on the data centers, networks and power the sector inherits (framework §1.5). It names loss of settlement finality within the operative window as the class's governing consequence (framework §3.2), and it leaves to this class the calibration of the settlement-finality window it proposes as the sector's consequence measure, together with its escalation structure (framework §3.6). This document takes both up. On the author's decisions of 14 October 2026 the class is written for large-value and wholesale settlement and the participant's interface to it in full, with retail payment systems stated at the depth of the sub-index for a later revision to develop (Section 2.2); the participant's payment-instruction endpoint is placed in this class and the financial messaging network is stated as a boundary whose home is left open (Section 8.3, Section 10.4); and the class proposes three reference events of its own (Section 9.2).

1.2 Scope

The class covers the systems that move value between institutions with finality, as the framework's register names them: the Fedwire and CHIPS large-value systems, the ACH network, the card networks and the digital payment platforms (framework §3.2, §8.2). For large-value and wholesale settlement it covers the whole path from the decision to pay to the moment the payment cannot be undone: the participant's decision and the instruction it composes; the participant's endpoint, where the instruction is authenticated and sent; the settlement system's acceptance, queuing and liquidity management; the act of settlement on the books of the settlement agent; and the confirmation that returns to the participant and on which it then acts. It includes the operator of the settlement system and its contingency arrangements, and the participant's own arrangements for paying when its usual path is lost.

Three neighbors are separate classes of this series. The exchanges, the central counterparties that clear trades and the securities settlement complex are *Market Infrastructure* (DBA-FIN-MK-

FC2). The deposit and account-ledger systems, the ATM networks and the online and mobile banking through which the public holds and moves money are *Core Banking and Ledger* (DBA-FIN-BK-FC3). The sector's coordination, through the information-sharing and data-vaulting mechanisms and the federal systemic-oversight function, is the coordination class, DBA-FIN-OpsCo. They are covered here only where settlement depends on them. Where a securities trade settles against a cash payment, the securities leg is the market-infrastructure class's and the cash leg is this class's; where a bank's customer orders a wire, the customer's account is the core-banking class's and the instruction the bank then sends to the settlement system is this class's.

Three boundaries with other series are drawn in the framework and are taken here as given (framework §1.2, §6.1). The data centers on which settlement systems and participants run are the data-center series'. The networks over which instructions travel, and the position, navigation and timing service on which their sequencing and time-stamping rest, are the communications series'. The power beneath both is the energy series'. A fourth is drawn with the series written since the framework: the software from which settlement systems and endpoints are built, and the trust plane that authenticates their messages, are the information-technology series' (*Software, Hardware and Product Supply Chain*, DBA-IT-SW-FC3; and its Connectivity and Protocols class). This class inherits each of them, and adds what the framework calls the systemic-consequence layer (framework §1.5). Section 8.3 states the boundaries.

What makes a system a member of this class is the same everywhere: it moves value between institutions and, at a stated moment, makes the movement final. The particular rules, participants and hours differ from system to system, and that detail belongs in each system's own design basis.

1.3 The Reference Systems

Three positions in the payment chain are used as reference cases. As in the first logical class of the corpus (*Software, Hardware and Product Supply Chain* §1.3), the reference cases of a logical class are positions rather than facilities, each defined by what it is trusted to do and by what follows when it fails.

The central-bank real-time gross settlement system settles each payment individually and finally on the books of the central bank, as the Federal Reserve's Fedwire Funds Service does in the United States and the Bank of England's RTGS system does for sterling. Its participants hold their settlement accounts in it, and the payment systems of the private sector settle their own positions through it. It bounds the availability chain, because when it stops, every payment that must settle in central-bank money that day waits, and the day it must settle on is fixed. The outage of the Bank of England's RTGS system on 20 October 2014, conferred by the framework as DBA-FIN-RE-2 (framework §9.1; Section 5.1), is the case.

The private large-value system settles the payments of its members by netting them against one another so that each moves only its net position, as CHIPS does for dollar payments. It is designated systemically important by the Financial Stability Oversight Council (Section 8.4) and

depends on the central-bank system to fund and close its day. It bounds the concentration of liquidity, because its members' payments to one another are held in its queue until they can be released, and a failure that holds the queue holds them all.

The participant's payment-instruction endpoint is the point at which a bank composes, authenticates and sends the instruction that a settlement system or a correspondent will act on, and receives the confirmation that tells it what happened. It sits inside the participant, outside the settlement system, and it is trusted by everything downstream because its messages authenticate. It bounds the integrity chain, because an adversary who controls it sends instructions that are, in every sense the system can test, the bank's own. The theft from Bangladesh Bank's account at the Federal Reserve Bank of New York in February 2016, conferred by the framework as DBA-FIN-RE-1 (framework §9.1; Section 5.5), is the case.

Retail payment systems, the automated clearing house, the card networks and the instant payment systems, are stated at the depth of the sub-index rather than as reference cases (Section 2.2).

1.4 Class Type

This is a logical facility class under the author's determination of 30 September 2026 (Facility-Class Types Are Physical, Coordination and Logical), which types DBA-FIN-PS-FC1 as logical: a system or service running on inherited physical infrastructure. It sits under the Payment and Settlement (PS) function class (framework §3.1, §3.2). It is the second logical class written in the corpus, after *Software, Hardware and Product Supply Chain*, and it follows that class's adaptation of the physical classes' structure: reference positions in place of reference facilities, parameters stated in time and value rather than in mass or pressure, and adequacy resting at the end on functions the class does not own (Section 6.7). Two things differ. This class's consequence is not a blast radius across hosts but a window that closes: settlement either completes before the deadline that gives it meaning or it fails, and the harm grows with the value left unsettled and the time it waits. And this class's governing act is a single decision with a single irreversible effect, the release of a payment into finality, which makes the placement of the Bright Line here unusually concrete (Section 7.3).

1.5 Relationship to Governing Documents

This instrument is issued under the Financial Services Sector Framework and inherits the DBA-MA parent framework. It does not repeat the framework's taxonomy, bounding principles, consequence measure or Bright Line and Command Broker architecture; it inherits them, and Section 9 is the required inheritance map. Where the framework is still a skeleton (framework §4, §5 and §7), this instrument develops the class-level content ahead of it and records in Section 9 what the framework should take up. Section references of the form framework §N resolve against the Financial Services Sector Framework in force.

Governing instruments. This instrument does not own the concepts it applies. The Bright Line — its placement criterion, its granularity, its scope of application and the obligation it creates — is defined in FD-BL (*The Bright Line*), which governs. The Broker role that discharges that obligation is defined in FD-BR (*The Broker*). Layer decomposition, the determinism boundary as a scope gate, and the machine-learning carve-out are defined in FD-LD (*Layer Decomposition and Determinism*). Local sovereignty is defined in FD-LS (*Local Sovereignty*). Two determinations bind under FD-BL: FD-BL-D1 (*Mode of Discharge*) governs how the authorization obligation may be discharged, and FD-BL-D2 (*Indication Integrity*) governs the integrity of the basis on which an authorization is granted. FD-EV (*Envelope*) reserves the term envelope. Placement is on the consequence of the individual action, assessed per action; this instrument applies that definition and does not vary it, and where its application to this class is unsettled the instrument says so (Section 7.3, Section 10.5). The Foundational Definitions are cited without a version token, and FD-REG (*Governing Stack Register*) is authoritative for their version state.

2. The Payment and Settlement Facility Class

2.1 Defining Characteristics

Five characteristics define the class, and each is a reason a payment system's failure reaches farther than the system.

The first is finality. A payment settled in the system cannot be undone by the system; that is what the system is for. Finality is what lets the receiving institution use the money at once, and it is what makes a payment on a false instruction a loss rather than an error. The same property that is the class's purpose is the reason its integrity consequence cannot be recovered after the moment of settlement.

The second is the deadline. Payments are owed on a value date, and many are owed by an hour within it: the close of a foreign-exchange settlement window, the completion of a house purchase, the funding of a clearing house, the payroll of an employer. A payment that settles late may be as harmful as one that does not settle, and the harm is set by the deadline it misses, not by the length of the delay.

The third is interdependence. Each participant pays out of what it has received, so that a payment one participant cannot make strands the next one's. Liquidity that is stuck in a queue, at a participant that cannot send, or at a system that has stopped, is liquidity every counterparty in the chain is short of. A failure at one point is felt as a shortage at every other.

The fourth is trust by authentication. A settlement system acts on an instruction because it authenticates as coming from the participant, and the participant acts on a confirmation because it authenticates as coming from the system. Authentication shows who sent a message. It does not show that the sender meant it.

The fifth is concentration. A few systems settle almost all the value, a few participants send most of it, and those participants depend on a few providers of software, messaging and hosting. A failure at any of the few is a failure across the many.

2.2 The Systemic-Importance Sub-Index

The framework sub-indexes the sector's classes by systemic importance, keyed to the Financial Stability Oversight Council's designation of systemically important financial market utilities and to the global systemically important bank designation (framework §2.2, §3.1, §8.2). This instrument states three tiers for the class.

The first tier is the systemic settlement system: a central-bank real-time gross settlement system, or a payment system designated systemically important, through which a large share of the value of a currency settles and on which other systems depend to close their own day. Its failure is a failure of the currency's settlement, and it carries the full design-basis load of this class.

The second tier is the systemic participant and its endpoint: a bank designated globally systemically important, or any participant whose payments into a first-tier system are large enough that its failure to send strands the liquidity of others, together with the endpoint from which it sends. The class treats the endpoint of a second-tier participant as carrying the design-basis load of the system it reaches, because what it sends settles there with finality.

The third tier is every other system and participant. It is in the class, and the class's provisions apply to it in proportion, but it does not carry the design-basis load of the first two.

Retail payment systems are stated at the depth of this sub-index only, on the author's decision of 14 October 2026. The automated clearing house, which settles batches of retail and payroll payments on deferred net terms; the card networks, which authorize purchases in seconds and settle them later; and the instant payment systems, which settle retail payments individually and finally at any hour, including the Federal Reserve's FedNow service, launched in July 2023, each carries the tiers above, and an instant payment system that settles in central-bank money at all hours may reach the first. Their design bases, and the public-facing consequence of a retail outage, are held for a later revision (Section 10.3).

2.3 Settlement Models and Range

The class spans several ways of reaching finality, and the design basis turns on which a system uses. A real-time gross settlement system settles each payment as it is released, in central-bank money, and holds a payment in a queue only while its sender lacks the funds. A deferred net settlement system accumulates payments and settles the net positions at set times, so that a payment is not final until the batch settles. A hybrid system offsets payments continuously and releases them in groups, saving liquidity at the cost of a queue whose contents are not yet final. And a payment-versus-payment system settles the two legs of a currency trade together, so that neither is final unless both are.

The range is wide. At one end is a central-bank system that settles a currency's interbank value each day; at the other a small participant that sends a few payments through a correspondent. The class is written to fit both, because a false instruction from the smallest participant settles with the same finality as one from the largest.

2.4 Consequence Context

The class's consequence is the loss of settlement finality within the operative window, which the framework names as the class's governing consequence (framework §3.2), and it takes two forms. The first is settlement that does not complete in time: the payment is valid, the system or the sender cannot settle it, and the harm grows with the value left unsettled and with the deadlines it misses. The second is settlement that completes on a false instruction: the system works, the payment is final, and the value is gone to a party who was never meant to have it.

Neither needs a physical failure, and both reach other sectors. On 20 October 2014 the Bank of England's RTGS system was unable to settle for about nine hours; the settlement day was extended from 16:20 to 20:00, 142,759 CHAPS payments worth £289.3 billion settled that day, and the independent review estimated that about seven hundred house purchases may not have completed until the next day or later (Deloitte, *Independent Review of RTGS Outage on 20 October 2014*, March 2015). In February 2016 thirty-five fraudulent instructions sent from Bangladesh Bank's own endpoint asked for the transfer of about $951 million from its account at the Federal Reserve Bank of New York; five were paid, $81 million to accounts in the Philippines and $20 million toward Sri Lanka, as reported, and most of the $81 million has not been recovered. The outward consequence the framework describes, interrupted payroll, cash distribution and commerce and, beyond a threshold, the erosion of confidence (framework §6.2), is the cascade both forms set off.

3. The Facility-Class Design Envelope

3.1 Why Payment and Settlement Is the First Class

Three reasons put this class first. The framework names the settlement-finality window as the sector's consequence measure and leaves its calibration to this class (framework §3.6), so the other two classes of the series cannot be sized until this one is written. The other classes depend on it: a securities trade is not settled until its cash leg is, and a bank's customers are not paid until the bank's payments settle. And the record is good: the events of this class are investigated by central banks and their independent reviewers, by courts and by the operators of the systems themselves, so the parameters can be stated against what is known.

3.2 The Two Consequence Chains

Every event in the class reaches the counterparties by one of two chains.

Chain F, the finality chain, begins with something that stops valid payments from settling: the settlement system fails, a participant cannot send, the network between them is lost, or the time on which the system sequences payments is wrong. Payments queue; liquidity is stranded; deadlines pass. Its consequence is measured by the value left unsettled and by the deadlines missed, and it grows with time. It is the availability chain of this class. The Bank of England's outage of 2014 is the case.

Chain V, the validity chain, begins with an instruction that is not what its sender intended: composed by an adversary at the participant's endpoint, altered between the participant and the system, or produced by a defect or a component acting on a false basis. The system authenticates it and settles it with finality, and the confirmation the participant then sees may itself be false. Its consequence is the value paid away, which finality makes irrecoverable through the system, and the confidence lost when it becomes known that an authenticated instruction can be false. It is the integrity chain of this class. The theft from Bangladesh Bank is the case.

The two chains meet at the moment of finality. Chain F is the failure to reach it; Chain V is reaching it on the wrong basis. A design basis that answers Chain F by restoring settlement as fast as it can will, if it restores on records it has not checked, open Chain V; and one that answers Chain V by checking each payment before finality will, if the check is slow, lengthen Chain F. The tension is real, and the class's parameters are written to state it rather than hide it (Section 3.3, Section 4.6).

3.3 Design Parameters at Class Resolution

Five parameters govern the envelope. Together they are this class's calibration of the settlement-finality window the framework proposes (framework §3.6), and Section 10.2 records them for the framework to take up.

The first is the finality window: for each payment the system settles, the interval between the moment it can be released and the latest moment its settlement still serves its purpose, set by its value date and by the deadline within the day that it must meet. The window is not one figure for a system. It is a distribution over the payments the system carries, and its shortest members, the foreign-exchange pay-ins, the funding of clearing houses and the payments that complete other settlements, set the system's requirement.

The second is the recovery time: the time from a disruption to the resumption of settlement. The international principles for financial market infrastructures ask that a system's critical information-technology systems be able to resume operations within two hours of a disruptive event and that it be able to complete settlement by the end of the day of the disruption (CPSS-IOSCO, *Principles for Financial Market Infrastructures*, 2012, Principle 17), and the cyber guidance of 2016 applies the two hours to the resumption of critical operations after a cyber event (CPMI-IOSCO, *Guidance on Cyber Resilience for Financial Market Infrastructures*, June 2016). The class takes the two hours as the reference point it is and not as a design-basis value, for the reason the third parameter states.

The third is the integrity verification time: the time needed to establish that the records on which settlement resumes are correct, after an event that may have altered them. When the Bank of England's system stopped in 2014, the concern was a possible loss of integrity, and settlement resumed only after a reconciliation that showed every account group's projected position matched (Deloitte review, paragraphs 41 and 183). The independent review recommended preparing for a loss-of-integrity scenario (paragraph 86). An operator responding to the international guidance in 2016 asked that it recognize that resumption may need to be balanced against preserving data and containing corruption (CLS Bank International, response to the consultative report, 23 February 2016). The class requires the verification time to be stated alongside the recovery time, and the recovery time to be stated as the time to resume on verified records. The verification time is prepared, not estimated. In 2014 the reconciliation the team estimated at 60 to 90 minutes took 165, in part because it was not at first known whether integrity had been lost (*The Outage of the Bank of England's RTGS System* F-1). A first-tier system therefore holds prepared and exercised reconciliations for each design-basis event of Section 5, beginning with the step that establishes whether integrity is in question, so that the verification time it states is a figure it has measured (*The Outage of the Bank of England's RTGS System* R-1). The tension is now recognized in the international guidance itself: the toolkit CPMI-IOSCO published in September 2026 states that resumption within two hours would likely be conditional on the integrity of data, and that an infrastructure resuming within two hours after an attack may be resuming to the same vulnerable state, as the instrument records it.

The fourth is the extension capacity: how far the system's day can be extended, and what that extension costs downstream. In 2014 the day was extended by three hours and forty minutes, and all submitted CHAPS payments settled that day; two early-closing Asian markets had their sterling pay-ins to the foreign-exchange settlement system delayed by about an hour and a half (Deloitte review, paragraph 17). Extension converts lost hours into missed downstream deadlines rather than into unsettled value, and the capacity to extend is bounded by the deadlines of the systems and markets that depend on the day's close.

The fifth is stranded value: the value of payments queued or unsent, integrated over the time they wait, and the number of participants whose own payments depend on receiving them. It is the measure of Chain F's consequence at any moment, and the measure against which the escalation of the next paragraph is set.

The framework asks for an escalation structure (framework §3.6), and the class proposes three zones. In Zone 1 settlement resumes within the recovery time on verified records, and the window is met for every payment; the event is operational. In Zone 2 settlement resumes within the value day, by extension if needed, and the window is met for most payments but missed for those with deadlines inside the day; the event is a settlement-system event, and the deadlines missed are stated. In Zone 3 settlement does not complete within the value day, or completes on records that cannot be verified; finality is lost for the day's payments or cannot be relied on, and the event is systemic. The design basis requires that the first-tier systems stay within Zone 2 for

every design-basis event of Section 5, and that the conditions under which they would enter Zone 3 be stated with the decision that would then be taken and the authority that would take it.

This instrument states the parameters and their governing logic. Their quantitative values for a system await validation against what operators and participants already hold: their settlement records, their incident and exercise reports, and their contingency rehearsals (Section 10.1). They are not asserted here as numbers, and the figures quoted are the events' own.

4. The Instruction and the Moment of Finality

4.1 Status and Use

Every event in this class reaches its consequence through an instruction that settles, or one that cannot. The path of the instruction from the participant's decision to the moment of finality is the one place the class can govern both chains, and this section states what the class requires of it, as the supply-chain class states its requirements of the update channel (*Software, Hardware and Product Supply Chain* §4).

4.2 Authentication Is Not Authorization

A settlement system authenticates an instruction to know who sent it. In February 2016 the fraudulent instructions were, on the Department of Justice's account, fraudulently authenticated messages sent from terminals that interfaced with the bank's messaging system, after the bank's network had been compromised; they authenticated as the bank's because they were sent from the bank's own endpoint. Nothing in the settlement system's authentication could tell them from the bank's real instructions, and it was not built to. The class therefore requires that what the participant intended be established separately from what its endpoint sent: authentication protects the channel against impersonation from outside, and protects against nothing done from inside the participant's own endpoint. This is the finding the supply-chain class makes of software signatures (*Software, Hardware and Product Supply Chain* §4.2), made here of payment messages.

4.3 Reconciling What Was Sent With What Was Intended

The participant learns what its endpoint did from the records the endpoint keeps and the confirmations it receives. In the Bangladesh Bank case the adversary's malware, as reported at the time by investigators and by Reuters and others, interfered with the records and the printed confirmations through which the bank would have seen the transfers, and the theft was timed across the bank's weekend, the New York weekend and a public holiday in the Philippines, so that each party's attention was absent when the next party acted. The bank's stop-payment request reached the receiving bank in the Philippines after much of the money had been withdrawn, as reported.

The class requires that a first- or second-tier participant reconcile, by a path independent of the endpoint that sent them, the instructions its endpoint sent with the payments it intended, at intervals short enough to stop a false instruction before its proceeds are beyond reach, and on every day the settlement system is open, whatever the participant's own working days. The confirmation it reconciles against is to be one it receives by a path the endpoint cannot alter, such as the settlement system's own statement of the participant's account. This is the indication-integrity obligation of FD-BL-D2 (*Indication Integrity*) applied to the participant's view of its own payments: a confirmation reported by the same endpoint that may have been compromised is not a basis for believing nothing is wrong. A failure of the endpoint's own reporting is a sign. In February 2016 the confirmations did not print and were taken for a printer fault, and the next day the interface software would not start; both were treated as faults to be fixed rather than as indications (*The Theft From Bangladesh Bank's Account* F-3). A missing confirmation, a record that does not match, or endpoint software that fails is therefore treated as an indication of compromise, and triggers the independent reconciliation of this section or a hold on the participant's instructions (*The Theft From Bangladesh Bank's Account* R-3).

4.4 The Receiving Side's Check

Most of the value asked for in February 2016 was not paid. As reported, thirty of the thirty-five instructions were not paid, an outcome the reporting states without saying why, and a routing bank queried the Sri Lankan transfer after the beneficiary's name was misspelled. The Federal Reserve Bank's query of 4 February reached Dhaka at about four in the morning of the bank's weekend. Checks on the receiving side, made by parties who saw only the instructions, held most of the value. The class records that defense and does not take credit for it: the checks that held were not designed against this attack, and the instructions they stopped were stopped for features an adversary can remove. The class requires that a first-tier system's checks on instructions it receives be stated as part of its design basis, with what they are designed to catch, and that a first-tier system hold the means to suspend a participant's instructions quickly on the participant's own report or on its own detection, at any hour it is open. The receiving system's own check stands before finality. A first-tier system holds for review, before release, any instruction that departs from the participant's established pattern by measures it states, and applies to its own release the release threshold the participant applies to its sending (*The Theft From Bangladesh Bank's Account* F-2, R-2). And a stop reaches every party through which the money can still move: the system, the participant and its correspondents hold a stop path that is answered whenever any of them can move the money, with its hours stated against all their calendars, because in February 2016 each party's weekend or holiday was the hour the next party acted (*The Theft From Bangladesh Bank's Account* F-1, R-1; noted to *Core Banking and Ledger*).

4.5 Contingency Settlement and the Switch to It

A settlement system holds a contingency arrangement for when its primary system cannot settle. In October 2014 the Bank of England's team chose to repair and restart the primary system rather than switch to its contingency system. The teams had a clear preference to resolve the problem within the primary system and judged there was time to reinstate it; the independent review endorsed fixing rather than switching (paragraph 81), for reasons it gave at paragraphs 48, 60, 63 and 78: switching would take as long as fixing, the contingency system lacked functionality and did not have the full resilience of the primary, and a switch could not be reversed until the weekend, the switch back mid-week never having been tested. In the same paragraph it said that early invocation of the contingency could have helped mitigate the immediate impact on the real economy, and at paragraph 60 it found that the decision was taken without a full understanding of that impact. It recommended reducing the barriers to switching (paragraph 86). A contingency that has not been exercised in the conditions in which it would be used weighs against itself in the moment of choice, and the design basis should not leave the choice to be made on that footing.

The class requires that the contingency arrangement of a first-tier system be exercised under the conditions of each design-basis event of Section 5, including a mid-week switch and the switch back, and that the decision to switch be one whose criteria and authority are stated in advance. It requires that a second-tier participant hold a means of sending its most critical payments when its usual endpoint or connectivity is lost, and that the settlement system accept them. Two things follow from the review's own reasons. The deciding authority is to know the deadlines before it decides: a first-tier system holds a map of its time-critical payments and their deadlines, including those of its non-direct participants and of the public, before the deciding authority from the first hour of a disruption (*The Outage of the Bank of England's RTGS System* F-2, R-2). And the decision to keep fixing expires: the switch criteria include the latest hour at which the contingency can still complete the day, and continuing to repair the primary past that hour is a named authority's decision to rely on the primary alone (*The Outage of the Bank of England's RTGS System* F-3, R-3).

4.6 Resuming on Verified Records

A system that has stopped may resume quickly or resume on records it has checked, and the 2014 outage shows the choice. The team ran a reconciliation and a controlled restart before reopening the system to its participants, with the system's liquidity-saving mechanism turned off and accounts reactivated in stages; phased settlement began at about 14:15 and settlement was fully resumed for all participants by 15:15, about nine hours after the error was detected (Deloitte review, paragraphs 43, 44, 183 and 188). Under a deliberate attack that alters records, the check is the whole of the defense, and a resumption that skips it settles whatever the attacker left. The class requires that a first-tier system state how it would establish that its records are correct after an event that may have altered them, how long that takes, and on what records it would resume if they could not be shown correct within the value day; and it requires that the decision to resume

on unverified records, if it is ever taken, be taken by a named authority who has been told what is not verified (Section 7.3).

5. Design Basis Events: The Envelope

The events below are the class's design-basis envelope, each stated with the event postulated, the consequence threshold, the function the class requires and the duration for which it is required. Sections 5.1 and 5.3 are non-adversarial; Sections 5.2 and 5.4 may be either; Sections 5.5 and 5.6 are adversarial and stand here because the validity chain is a governing chain of the class in every domain, not only under military action; and Section 5.7 is held at the depth of the sub-index. Inclusion is by consequence severity, not likelihood (framework §4.1).

5.1 Settlement System Outage Beyond the Recovery Time

Postulated event a first-tier settlement system stops settling, through a defect, a failed change, the loss of its processing site or its operator's error, for longer than its recovery time, on a day when payments with deadlines inside the day are due.

Consequence threshold payments that cannot settle within their finality window, stranded value across every participant, and deadlines missed downstream (Chain F); entry into Zone 2, and into Zone 3 if settlement cannot complete within the value day.

Required function the system resumes settlement on verified records within its stated recovery time, or switches to a contingency arrangement exercised under the same conditions (Section 4.5); the settlement day can be extended to complete every payment within the value day, with the downstream deadlines that extension breaches stated in advance; and the participants' most critical payments, identified in advance, are settled first. The deadlines are before the deciding authority from the first hour, and the switch criteria state the hour after which the contingency can no longer complete the day (Section 4.5).

Design-basis duration from the stop until every payment due that day is settled; for the extension, the latest hour downstream deadlines allow.

The case is the outage of the Bank of England's RTGS system on 20 October 2014, conferred by the framework as DBA-FIN-RE-2 (framework §9.1), whose instrument reads it as non-adversarial. The system opened at 06:00 and the error was detected at 06:02. A change made over the preceding weekend to move one bank's CHAPS membership to another, which looked routine but was not, set off a previously undetected design defect in one process of the system, which in turn set off two latent defects in another; the defects had been introduced by changes made in 2013 and 2014, and testing had not caught them (Deloitte review, paragraphs 23 to 27). Settlement resumed in phases from about 14:15 and fully by 15:15, the day was extended to 20:00, and all submitted CHAPS payments settled that day (paragraphs 9, 10, 43 and 44). The design-basis reading is that a first-tier system stayed within Zone 2 by extension and reconciliation, that the cost of the extension fell on the deadlines downstream of the system, and

that the contingency system, whose switch-back had not been exercised mid-week, was not the operator's preferred path, a decision the review endorsed while recommending that the barriers to switching be reduced, and while finding that early invocation could have helped the real economy and that the decision was taken without a full understanding of its impact (paragraphs 60 and 81).

5.2 Loss of Integrity in the Settlement Records

Postulated event an event, accidental or deliberate, after which the operator of a first-tier system cannot show that its records of accounts, balances and queued payments are correct.

Consequence threshold settlement that cannot resume without the risk of settling on false records (Chain V opened inside the system) or cannot resume at all within the value day (Chain F, Zone 3).

Required function the operator holds records from which correct positions can be re-established, protected so that the same event cannot reach them; states the integrity verification time of Section 3.3; and holds an arrangement, agreed with its participants, for completing the day's settlement or for carrying it to the next day on positions all parties accept.

Design-basis duration until settlement resumes on verified records or the agreed arrangement completes the day.

This event is developed here from the 2014 review's recommendation that the Bank prepare for a loss-of-integrity scenario (Deloitte review, paragraph 86), and from the international guidance's account of the tension between resuming quickly and resuming correctly. The sector's data-vaulting standard, Sheltered Harbor, addresses the corresponding problem for an institution's customer accounts, which is the core-banking class's (Section 8.3).

5.3 Loss of a Participant's Endpoint or Connectivity

Postulated event a first- or second-tier participant cannot send or receive instructions, because its endpoint, its connectivity or its processing site is lost, while it remains solvent and its counterparties expect its payments.

Consequence threshold the participant becomes a liquidity sink: it receives payments it cannot pass on, and every participant that expected its payments is short (Chain F).

Required function the participant holds a means of sending its most critical payments without its usual endpoint, which the settlement system accepts and which is exercised (Section 4.5); and the system's liquidity-saving arrangements are designed so that one participant's silence does not hold the queue for all.

Design-basis duration until the participant's usual path is restored or its payments are being made by the alternative.

The ransomware attack on a clearing member of the United States Treasury market in November 2023 shows this event at the boundary with *Market Infrastructure*, and is held for that class (Section 9.2).

5.4 Loss or Corruption of Time and Sequence

Postulated event the time on which a system sequences, time-stamps or closes its settlement cycles is lost or wrong, through the failure or denial of the timing service the system relies on.

Consequence threshold payments settled in the wrong order, cycles closed at the wrong time, or a system that stops because it cannot establish the time (Chain F; Chain V where the order of settlement changes who is paid).

Required function a first-tier system keeps its own time for the duration of its longest settlement cycle without the external service, and detects a time it is given that disagrees with its own.

Design-basis duration the longest interval the timing service can be denied, as the communications series states it.

The framework names the position, navigation and timing service as an acute dependency of the sector (framework §6.1). The communications series governs the service and its holdover (*Positioning, Navigation and Timing*, DBA-COMM-PN-FC4); this class states what a settlement system must be able to do without it.

5.5 Fraudulent Instruction Through a Participant's Endpoint

Postulated event an adversary gains control of a participant's payment-instruction endpoint and sends instructions that authenticate as the participant's own, while concealing them from the participant's view of its payments.

Consequence threshold value paid with finality to parties of the adversary's choosing, beyond recovery through the settlement system (Chain V).

Required function the participant reconciles what was sent against what was intended by an independent path, on every day the system is open (Section 4.3); an instruction from an in-scope component whose consequence reaches the threshold, as the participant's release threshold implements it, releases only on the Command Broker's authorization, given on a basis corroborated by a path the endpoint cannot alter (Section 7.3); the receiving system's checks are part of its stated design basis and it can suspend a participant's instructions at any hour it is open (Section 4.4). A failure of the endpoint's reporting triggers the reconciliation or a hold (Section 4.3); the receiving system holds for review any instruction that departs from the participant's pattern; and a stop reaches every party at every hour any of them can move the money (Section 4.4).

Design-basis duration from the first false instruction until the participant's endpoint is cleared and every false instruction is stopped or traced.

The case is the theft from Bangladesh Bank's account at the Federal Reserve Bank of New York of 4 and 5 February 2016, conferred by the framework as DBA-FIN-RE-1 (framework §9.1). On the author's ruling of 14 October 2026 that a state's covert theft operations are military actions, its instrument records it as military action without the military-action tier, denial not being present (*Design Basis Layers* §6.4). The United States Department of Justice states that the bank's network was compromised by spear-phishing, that its terminals interfacing with the messaging system were accessed, and that fraudulently authenticated messages directed the Federal Reserve Bank of New York to transfer funds out of Bangladesh, $81 million of which was stolen; the Department attributes the theft to a group it links to the government of North Korea (Department of Justice, 6 September 2018). The attribution is the United States government's, and this instrument makes no finding of attribution. The figures beyond the Department's are as reported at the time by Reuters and others: thirty-five instructions for about $951 million, five paid for about $101 million, and the Sri Lankan transfer of $20 million reversed; thirty of the thirty-five were not paid, an outcome the reporting gives without saying why. The design-basis reading is that a central bank's authenticated endpoint sent instructions it did not intend, that the participant could not see them in time because its own view was the view the adversary controlled, and that the value held was held by checks on the receiving side that were not designed against the attack.

5.6 Compromise of the Settlement System's Software or Operator

Postulated event an adversary reaches the settlement system itself, through its software supply chain, its operator's network or an insider, and alters how it settles, what it records or what it reports.

Consequence threshold settlement on false records or false rules across every participant (Chain V), or a stop the operator cannot safely reverse (Chain F, Zone 3).

Required function the requirements of *Software, Hardware and Product Supply Chain* §4 for the system's software as a first-tier product; changes to the system's settlement logic, limits or participant set only through a governed act in the sense of *Local Sovereignty* §3.2; and the records and arrangement of Section 5.2.

Design-basis duration the dwell of the compromise, and thereafter until settlement resumes on verified records.

No reference event is proposed for this category; the class cites the supply-chain class's events for the path. The 2014 outage shows that a change whose routine appearance hid a non-routine effect can stop a first-tier system without any adversary (Section 5.1), and the design-basis reading is that a change an adversary makes will be designed to look routine.

5.7 Retail Payment System Failure

Postulated event a card network, automated clearing house or instant payment system fails, so that purchases cannot be authorized or retail and payroll payments cannot settle.

Consequence threshold at the depth of the sub-index only; to be developed (Section 10.3).

Required function at the depth of the sub-index: a retail system that settles in central-bank money at all hours carries the requirements of Sections 5.1 to 5.6 as a first-tier system; the public-facing consequence is held for a later revision.

Design-basis duration to be developed.

Table 5-1. Facility-class design-basis event categories.

Event category	Postulated event (class-specific)	Consequence chain	Framework source
Settlement system outage beyond the recovery time	First-tier system stops; deadlines inside the value day at risk	Chain F	§4.1 candidate (settlement-system outage across the finality window)
Loss of integrity in the settlement records	Operator cannot show its records are correct	Chain V + Chain F	developed here
Loss of a participant's endpoint or connectivity	Solvent participant cannot send; becomes a liquidity sink	Chain F	§4.1 candidate (inherited failures); §6.1
Loss or corruption of time and sequence	Timing service lost or wrong	Chain F (Chain V by order)	§6.1 (position, navigation and timing); §7
Fraudulent instruction through a participant's endpoint	Authenticated instructions the participant did not intend	Chain V	developed here; §5
Compromise of the settlement system's software or operator	System's settlement, records or reports altered	Chain V + Chain F	developed here; DBA-IT-SW-FC3
Retail payment system failure	Card, ACH or instant payment system fails	Chain F	§3.2; sub-index depth

6. Military-Action Extension: DBA-MA for Payment and Settlement

6.1 MA Extension Applicability Determination

Every first- and second-tier system and participant in the class performs the applicability determination the series makes mandatory, under the two-factor test of geolocation and strategic value, with strategic value driven by systemic-importance designation (framework §7). The framework calls financial infrastructure a primary objective of economic warfare and the settlement backbone the highest-strategic-value node (framework §7). For a logical class the first factor is restated as the supply-chain class restates it (*Software, Hardware and Product Supply Chain* §6.1): where the system's processing sites and their contingency sites are; where its participants and their endpoints are; and under whose law the system, its messaging and its providers operate, which sets whether a government can compel a participant's exclusion. The determination is recorded, it is time-varying and re-triggered on the indicators the DBA-MA parent framework defines, and any scope reduction carries a recorded basis (Section 6.9).

6.2 The Four Characteristics Applied to the Class

Applied to payment and settlement, the four military-action characteristics take specific form, in the canonical statement of FD-DBL (*Design Basis Layers*), at its §4.4. Intentionality: the adversary chooses the settlement system, or a participant's endpoint, because the economy it serves stops when settlement stops, or because value taken from it funds the adversary. Persistence: the adversary holds a participant's network or endpoint across the time needed to learn its procedures and choose its moment, as the Bangladesh Bank theft required, the receiving accounts having been opened, as reported, many months before. Precision: the adversary chooses the hour, the day and the holidays across which each party's attention is absent, as that theft was timed across two weekends and a holiday, or chooses the day of a settlement deadline that cannot move. Denial: the adversary takes the power, communications and timing on which settlement depends, or strikes the processing sites, so that the system cannot settle and its participants cannot send, when the economy most needs to pay and be paid.

Under the author's rulings of 10 and 13 October 2026, the military-action tier is not met without denial, and all four characteristics must be established on the record; a characteristic not determined counts as not present (*Design Basis Layers* §6.4). On the author's ruling of 14 October 2026 a state's surveillance and covert theft operations, the sale of drugs and the laundering of money among them, are military actions (*Design Basis Layers* §6.4), so a state's theft that shows intentionality, persistence and precision and does not deny the system is recorded as military action that realizes the scenario it realizes, without the tier, as DBA-FIN-RE-1 records Bangladesh Bank. On a further ruling of the same date, the four characteristics may be read against a campaign on a class's dependencies rather than against the facility as an aim, the elements precision aims at in *Design Basis Layers* §4.4 not needing to belong to the facility under analysis; DBA-FIN-RE-3 reads Ukraine that way. The tier of each reference event is its instrument's to determine.

6.3 Settlement Under a Campaign on the Grid and Communications as the Governing MA Scenario

The governing military-action scenario is a campaign that strikes the power and communications of a country repeatedly, over months, while the country's economy must continue to pay its soldiers, its pensioners, its suppliers and its imports. Its settlement system must settle every day it is needed, its participants must reach it, and the public must be able to pay and be paid, while the grid fails for hours at a time and the networks with it. No credit is taken for the grid, the public networks or the timing service, or for processing sites inside the country that the campaign can reach.

The extension requires three things. It requires that a first-tier system settle through the campaign, from processing sites and contingency arrangements that do not share the grid, the networks or the exposure that the campaign targets, with power, communications and time of their own for the longest interruption the campaign can impose. It requires that its second-tier participants reach it by paths that survive the same campaign. And it requires that the integrity of the system's records be established at each resumption, because a campaign that denies the system will also try to alter it (Section 6.5). Two further requirements come from the event. The Assured Finality Condition reaches the edge: a first-tier system's design basis states how the public will pay and be paid through the campaign, with enough access points holding their own power, communications and cash for the campaign's longest outage, their number and distribution stated in advance (*Ukraine's Payment System Through the War* F-1, R-1; noted to *Core Banking and Ledger* and to the retail revision of Section 10.3). And the permissions continuity needs are granted before the war: where the applicability determination is positive, the authorities grant or pre-authorize, with conditions, the permissions continuity depends on, the location of data, services from abroad and the days and hours of settlement among them, and the system exercises them, because in 2022 the plans of 2014 were ready and the permission for banks to use cloud services abroad came in the first weeks of the war (*Ukraine's Payment System Through the War* F-2, R-2).

The empirical case is the payment system of Ukraine since February 2022, conferred by the framework as DBA-FIN-RE-3 (framework §9.1), whose instrument finds the military-action tier met, all four characteristics established on the record. On the account of the National Bank of Ukraine's director for payment systems, its System of Electronic Payments, which carries about 97 percent of hryvnia interbank transfers on the bank's release for 2022 and more than 98 percent on the interview of 2023, has operated since the full-scale invasion at the same level of reliability as before it; the bank activated continuity plans drawn up in 2014, allowed banks to use cloud services hosted outside the country so that data and operations could survive the destruction of facilities within it, and, when strikes on the grid began in the autumn of 2022, organized a network of bank branches with their own power and communications to serve the public through long blackouts, announced with more than 1,000 branches and numbering more than 2,370 by July 2023 (European Payments Council, interview with the National Bank of

Ukraine, 28 July 2023; the National Bank's own releases and its deputy governor's article of 13 May 2022). The design-basis reading is that a first-tier system was kept settling through a campaign of the kind this scenario postulates, by arrangements prepared years before it, and that the public's access to payment was kept by provision at the edge as well as at the center.

6.4 Exclusion as Denial

The second military-action scenario is denial by exclusion. A payment system, a messaging network or a correspondent may be closed to a country's participants by the decision of a government, as the European Union required in March 2022 when it barred the provision of specialized financial messaging services to seven Russian banks (Council Regulation (EU) 2022/345). This instrument takes no position on any such decision. It records that exclusion denies a participant settlement as surely as a strike on its processing site, and that a design basis for a participant whose government may be in conflict with the governments of the systems it relies on states what it would do if they were closed to it. The class requires such a participant's design basis to state the systems and networks its critical payments depend on, the law under which each operates, and the path by which it would make those payments if any were closed.

6.5 The Validity Attack Timed to a Crisis

The third scenario joins the two chains. An adversary holding a participant's endpoint, or the settlement system's own records, acts not to steal but to corrupt: it sends or alters payments so that, when the crisis it has chosen arrives, the system's records cannot be trusted and settlement cannot resume without a verification that takes longer than the economy can wait. The Bangladesh Bank theft is the peacetime form of the access; the crisis is the moment the extension postulates. Against it the class relies on the independent reconciliation of Section 4.3, the verified resumption of Section 4.6, the protected records of Section 5.2, and the Command Broker's authorization of the payments whose consequence requires it (Section 7.3).

6.6 Cross-Sector Interaction

Under military action the class is tied to every sector going out and to five coming in. Going out, the loss of settlement stops payroll, the purchase of fuel, food and medicine, the funding of hospitals and emergency services, and the settlement of the securities and foreign-exchange markets; and beyond a threshold, as the framework says, it erodes the public confidence on which the monetary system rests (framework §6.2). Coming in, the class depends on electric power, the data centers, the communications networks and their timing service, and the software and trust plane of the information-technology series (framework §6.1). An adversary need not attack a settlement system to stop it; an attack on its power, its networks, its time or the software it runs on, timed to a deadline that cannot move, is enough.

6.7 The Adequacy Result: Assured Finality Condition

The test of whether the class's protection is adequate is whether a first-tier system and its second-tier participants can reach what this document calls the Assured Finality Condition. When the grid, the public networks, the external timing service and any one processing site are lost, and when an adversary may have altered a participant's endpoint or the system's own records, the system settles the day's critical payments within their finality windows, on instructions whose validity has been established before finality, from records whose correctness has been shown, and from sites and paths whose power, communications and time do not depend on what the event removed. The condition bounds Chain F by the recovery time, the extension capacity and the contingency arrangement, and bounds Chain V by the gate before finality and the verification of records.

The class has few protective functions in the sense of *Local Sovereignty* §3.2, because a settlement system's protection is not a measurement against a setpoint executed on physical law. What it has in their place are deterministic controls on the path to finality: the limits on a participant's payments, the release thresholds above which an instruction waits for authorization, the suspension of a participant, and the end-of-day close. The class requires that these change only through a governed act in the sense of *Local Sovereignty* §3.2, by whatever component a change is proposed. That is a requirement of the class's design basis, not a placement; where an in-scope component proposes such a change, the change is also an action placed on its own consequence (*The Bright Line* §4.3). The power, cooling and fire protection of the processing sites are the data-center series' and are inherited (framework §6.1).

The end state may also be reached by deferral. Carrying the day's unsettled payments to the next day, on positions all parties accept, ends the system's obligation for that day at the cost of every deadline inside it. Like the evacuation of a hospital or the disconnection of a compromised product, it is a protective action with an authority named in advance, a trigger stated in advance, and a statement of what it costs and to whom.

6.8 Adequacy and the Declared Residual

In the non-adversarial domain, the Assured Finality Condition is achievable with the Section 4 and Section 5 requirements: a contingency arrangement exercised under the conditions it would be used in, an extension capacity stated against the deadlines downstream, a reconciliation independent of the endpoint, and records from which correct positions can be re-established. Against a deliberate adversary, adequacy has limits.

Three residuals are declared, in the output form of *Defense in Depth* §3.5. First, a false instruction that passes the participant's own authorization, because the adversary has deceived or coerced the person who authorizes, settles with finality; beyond the design act the residual is the value paid, and the response is the receiving side's checks and the recovery the law allows, not the settlement system. Second, a campaign that outlasts the endurance of the system's own power, communications and time ends its settlement; beyond that point the residual is deferral,

and the design response is an endurance sized for the campaign rather than for a storm. Third, exclusion by a government acts with the authority of the system that excludes; no arrangement inside the participant holds against it, and the residual is the participant's critical payments, answered only by the alternative paths of Section 6.4. These residuals bound the class adequacy claim.

6.9 Consequence-Basis Conditionality Check

Any relaxation of a conservative class parameter, such as a recovery time stated without the integrity verification time, an extension credited without stating the downstream deadlines it breaches, a contingency arrangement credited that has not been exercised mid-week, a release threshold set for convenience rather than for the consequence of the payments above it, or a participant's alternative path credited that the same event can deny, must be justified against the class baseline, and the larger the relaxation the stronger the analytical basis required. A relaxation without a recorded basis is not consistent with the class design basis. The framework does not yet state this principle for the sector, and Sections 9.1 and 10.10 record it for the framework to adopt.

7. The Dual Design Basis, Stated in Full

7.1 The Two Design Bases

The class carries two design bases at once: a settlement-integrity design basis, defined in this instrument, and an AI-governance design basis, governed by the Foundational Definitions and the Command Broker family and named and referenced here (framework §5). In a physical class the first is the physical design basis of the plant; in this class, as in the supply-chain class, it is the set of deterministic controls on the path to the class's irreversible act, here the moment of finality. The two design bases interact at the point developed in Section 7.4.

7.2 The Settlement-Integrity Design Basis: The Determinism Posture

The settlement-integrity design basis is the set of controls the class relies on to keep settlement timely and correct: the contingency arrangement and the switch to it (Section 4.5), the extension of the day, the verified resumption (Section 4.6), the independent reconciliation of what was sent with what was intended (Section 4.3), the receiving side's checks (Section 4.4), and the limits, thresholds and suspension that sit on the path to finality (Section 6.7). Most are deterministic functions.

Whether a function is deterministic decides whether the authorization rules apply to it at all. It does not decide where its actions fall on the Bright Line. Components that behave predictably within a stated range of inputs, and that have been formally tested in the deployed configuration, are outside the Command Broker obligation for the actions they take while their inputs stay in that range (*Layer Decomposition and Determinism* §4.3). They are not below the line; they are

outside the obligation only because, and only while, their behavior is bounded by that showing (*The Bright Line* §4.5).

The 2014 outage shows how that showing fails in this class. The settlement system was deterministic. The change made over the weekend put it into a configuration, a membership with one participant added and one deleted, that its tests had not exercised, and the defects it set off had been there since changes made in 2013 and 2014 (Deloitte review, paragraphs 23 to 27). The author determined on 8 October 2026 that the carve-out does not cover a carved-out function while it runs a configuration other than the one that was formally tested, including a configuration altered outside a governed act (Determination of 8 October 2026, A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out; for *Layer Decomposition and Determinism* §4.3 at its next revision). The class reads the participant set, the limits and the settlement rules as part of a settlement system's configuration, and a system that wishes to rely on the carve-out for its settlement logic must test each configuration a change will produce, in the form in which it will run, before the change takes effect. The review's recommendation that non-routine changes be deferred unless there is a compelling reason, and that testing be improved with separate test and pre-production environments (paragraph 83), is the operator's practice that the determination states as principle.

7.3 The AI-Governance Design Basis: Named and Referenced

Machine-learning and generative-AI systems are present throughout the payment chain: fraud and anomaly detection that scores each payment, sanctions screening that matches names against lists, liquidity-management systems that decide which payments to release and when, treasury agents that compose and submit payment instructions, and assistants that help operators investigate and repair. The framework names automated settlement and treasury operations among them and places the Bright Line for any action of a component within the scope of *Layer Decomposition and Determinism* §4.3 whose incorrect performance could settle a transaction irrevocably at or above the systemic-consequence threshold (framework §5). These systems are governed by the Bright Line and Command Broker architecture (framework §5) and by the Command Broker family, whose parent is CB-Framework_Cross_Sector (*Command Broker Framework*); CB-REG (*Command Broker Family Register*) is authoritative for the family's version state. The coordination class, OpsCo, is the locus at which the sector's placement and obligation are implemented (framework §3.5). Placement is by the consequence of each action, not by the component (*The Bright Line* §4.2, §4.3).

In this class the action is unusually concrete. The irreversible change a payment system makes is the release of a payment into finality, and the consequence of that release is set by its value, its recipient and the deadline it serves. Four cases illustrate the application, and none of them is a placement by component. A treasury agent within scope that composes and submits a payment instruction takes an action with each instruction, and the instruction together with its release is one action placed on its consequence: an instruction whose consequence can reach the threshold is above the line and requires the authorization of a qualified human, the Command Broker,

given on a basis the agent cannot alter. Payments within a range whose limits a qualified authority has approved in advance, the routine settlement of a participant's standing obligations, may be discharged in advance under FD-BL-D1 (*Mode of Discharge*), where the limits form a pre-approved envelope and the gatekeeper that enforces them, the release threshold, is verified; a change to the threshold is excluded from advance discharge (FD-BL-D1 R3). A liquidity-management system that decides the order in which queued payments are released takes an action with each decision, placed on the consequence of the delay it imposes on the payments it holds back. And a fraud or sanctions system that stops a payment takes an action placed on the consequence of the stop, which can be as severe as the consequence of the payment, because a stopped payment can miss a deadline that cannot move.

Whether *The Bright Line* §4.3, which defines one action for the industrial Line as one change to a process parameter, to a protective function's settings, trip logic, bypass state or actuation criteria, or to the pre-authorized envelope or gatekeeper, reaches a payment instruction was joined at v0.1 to the question the supply-chain class referred for a software release. The author decided the release case on 14 October 2026 (the author's determination of 14 October 2026 on vendor releases and staged releases (*Three Author Decisions: Continuity Classes, Vendor Releases and Staged Releases*)): *The Bright Line* §4.3 is not extended to a vendor's release to a population of hosts, and a staged release is placed on its whole planned reach with each widening its own action. The decision does not reach a payment instruction, which is a participant's act at its own endpoint. The class's gate before finality rests on the framework's own placement of the line for actions that settle a transaction irrevocably (framework §5), and that placement stands. Whether *The Bright Line* §4.3 should itself be widened for the logical classes is not decided; the class does not need it decided to apply the framework's placement, and states that it would put the question again only if a case arose that the framework's placement did not reach (Section 10.5).

Two further points apply whatever the answer. Any output on which an authorization is grounded, the record of what the participant intended, the confirmation of what was sent, a fraud score, carries the indication-integrity obligation of *Indication Integrity*: in February 2016 the bank's view of its own payments was the view the adversary controlled, and an authorization granted on that view would not have been substantive. The Command Broker must authorize on a basis corroborated by a path the endpoint cannot alter. And the decision to resume settlement on records that are not verified (Section 4.6) is itself an action of the highest consequence the class can produce, and is to be taken by a named authority on a basis whose limits are stated to it, whatever component proposes it.

7.4 The Interaction Point: The Gate Before Finality

The two design bases meet at the moment before a payment becomes final. The settlement-integrity design basis guarantees, so far as it can, that the system can settle on time and on correct records. The AI-governance design basis guarantees that no proposing system can release a payment into finality, or decide that settlement resumes on unverified records, without a

qualified human authorization the architecture can enforce, where the action's consequence requires it. The settlement system must be architecturally incapable of executing an above-line release originating from a component within the scope of *Layer Decomposition and Determinism* §4.3 without that authorization. In most of the corpus the Bright Line is drawn around a change to a physical process that can, in principle, be reversed. Here it is drawn around a moment after which nothing can be reversed, and that is why the gate stands before finality and not after it.

7.5 Scope Boundary

This instrument states the settlement-integrity design basis in full and names the AI-governance design basis without restating it. The Foundational Definitions own the Bright Line, its scope and its obligation; the Command Broker family owns the Command Broker; the coordination class owns the implementation of placement in the sector; and the market-infrastructure and core-banking classes own the placement of actions in their functions. This instrument applies them to payment and settlement and, where their application is unsettled, says so.

8. Cross-Sector Dependencies and Regulatory Context

8.1 What Other Sectors Depend On: Settlement

Every sector is paid through this class. Payroll, pensions and benefits reach households through it; the purchase of fuel, food, medicine and equipment by every operator in the corpus settles through it; the securities and foreign-exchange markets cannot settle their cash legs without it; and the government's own receipts and payments pass through it. The October 2014 outage showed the reach in miniature: house purchases delayed, foreign-exchange pay-ins late in Asia, and the settlement day of a currency extended by nearly four hours. The consequence in each consuming sector is that sector's to state; what this class states is the window within which settlement must complete for them.

8.2 What the Class Depends On

For a consumer sector the inward view is the defining one (framework §6.1). The class depends on the data-center series for the processing sites of systems and participants and their contingency sites; on the communications series for the networks over which instructions travel and for the timing service on which sequencing rests; on the energy series for power; and on the information-technology series for the software of the systems and endpoints and for the trust plane that authenticates their messages. Under the framework's reversed elevation, a data-center or communications asset that is the sole substrate for a first-tier system inherits that system's systemic-importance tier (framework §6.3). This class confirms the rule and adds that the substrate of a second-tier participant's endpoint carries the same elevation, because what the endpoint sends settles in the first-tier system.

8.3 The Boundaries with the Partner Classes and Series

Six boundaries are drawn. With *Market Infrastructure* (DBA-FIN-MK-FC2): where a securities trade settles against a payment, the securities leg and the clearing house are that class's and the cash leg is this class's; the ransomware attack on a Treasury-market clearing member in November 2023 is held for that class (Section 9.2). With *Core Banking and Ledger* (DBA-FIN-BK-FC3): the customer's account and the bank's ledger are that class's, and the instruction the bank sends to settle is this class's; the denial-of-service attacks on banks' online services of 2012 and 2013 are held for that class (Section 9.2). With the coordination class, OpsCo: the information-sharing and data-vaulting mechanisms and the systemic-oversight function are its members. With the financial messaging network: the participant's endpoint, where an instruction is composed, authenticated and sent, is this class's, on the author's decision of 14 October 2026, because what it sends is what settles; the messaging network that carries instructions between participants and to correspondents is stated as a boundary, and whether it belongs to this class, to the communications series or to the information-technology series is left open (Section 10.4). With the information-technology series: the software of systems and endpoints and its supply chain are *Software, Hardware and Product Supply Chain*'s, and the certificates and keys that authenticate messages rest on its Connectivity and Protocols class. With the data-center, communications and energy series: the substrate is theirs and is inherited.

8.4 Regulatory and Institutional Context

The class is governed by an international standard and by national regimes that apply it. The *Principles for Financial Market Infrastructures* of the Committee on Payment and Settlement Systems and the International Organization of Securities Commissions (April 2012) set the standard for systemically important payment systems: Principle 8 requires final settlement no later than the end of the value date, and Principle 17 sets the expectation for operational risk, including the aim to resume operations within two hours and complete settlement by the end of the day of a disruption. The *Guidance on Cyber Resilience for Financial Market Infrastructures* (June 2016) extends the second to cyber events. In the United States, Title VIII of the Dodd-Frank Act gives the Financial Stability Oversight Council the power to designate systemically important financial market utilities, which it used in July 2012 for eight utilities including CHIPS and CLS Bank, and the Federal Reserve's Regulation HH (12 CFR Part 234) sets risk-management standards for those it supervises. The Fedwire Funds Service operates under the Federal Reserve's Regulation J and its operating circulars, with finality as those rules provide. The messaging network's Customer Security Programme, launched in 2016 after the Bangladesh Bank theft, has since 2017 set mandatory security controls for its users' endpoints and required annual attestation against them. In Europe, the Settlement Finality Directive (98/26/EC) protects the finality of settlement in designated systems, and the Digital Operational Resilience Act (Regulation (EU) 2022/2554), applying from January 2025, sets operational-resilience requirements for financial entities and their critical technology providers.

The regulatory gap is the framework's (framework §2.3). The regimes state outcomes and practices for each system and institution: a two-hour aim, settlement by end of day, controls on the endpoint, testing and recovery. None states the set of events the payment system as a whole must be shown to withstand, expressed as a finality window against a postulated event; none resolves the tension between resuming in two hours and resuming on verified records, which an operator raised in responding to the guidance in 2016; and none treats a participant's endpoint as carrying the design-basis load of the system its instructions settle in. This class states them as design-basis requirements for the tiers that carry the load.

9. Inheritance Map and Anticipated Derivatives

9.1 The Inheritance Map

The series requires every facility-class instrument to state, at first publication, which elements it inherits unchanged, which it inherits with adaptation, and which it defines locally. Facility-class instruments inherit the framework's methodology and architecture and add their financial-function-specific consequence chain, the settlement-finality requirement and the recovery architecture (framework §8.1). Table 9-1 is the map for this class. Where the framework is a skeleton, the element is developed here and recorded for the framework to take up.

Table 9-1. Inheritance map for DBA-FIN-PS-FC1.

Element	Source in the framework	Inheritance mode
Function-primary placement, PS function class	§1.3, §3.1, §3.2	Inherited unchanged
Logical class type; consumer-sector shape	§1.5, §3.1; determination of 30 September 2026	Inherited unchanged; second logical class written (Section 1.4)
Systemic-importance sub-index	§2.2, §3.1, §8.2	Inherited with adaptation; three tiers stated, the participant's endpoint carried with the system it reaches, retail at sub-index depth (Section 2.2)
Design-basis principles (consequence severity, not likelihood)	§4.1	Inherited unchanged
Settlement-finality window and its escalation	§3.6 (proposed)	Calibrated here as five parameters and three zones (Section 3.3); for the framework to take up
Event categories	§4.1 (candidates)	Developed here ahead of the framework; for the framework to take up

Element	Source in the framework	Inheritance mode
Bright Line and Command Broker architecture	§5; FD-BL; Command Broker family	Inherited with adaptation: the gate before finality (Sections 7.3, 7.4); the joined question answered for the release case on 14 October 2026, the framework's placement standing for the payment
The instruction and the moment of finality	§3.2	Locally defined (Section 4)
Assured Finality Condition	§7 (skeleton)	Locally defined (Section 6.7)
Declared residuals and conditionality check	not yet in the framework	Locally defined; recorded for the framework (Sections 6.8, 6.9)
Consumer-sector inheritance and reversed elevation	§1.5, §6.1, §6.3	Inherited and extended to the participant's endpoint (Section 8.2)
Military-action applicability	§7	Inherited with adaptation: geolocation restated as for the supply-chain class (Section 6.1)

9.2 Anticipated Derivatives

System-, operator- and participant-specific design bases inherit this class document by reference and add only what is locally specific, each carrying a recorded applicability determination for any scope it reduces. The three reference events this class proposed at v0.1, on the author's decision of 14 October 2026, are conferred by the framework at §9.1: DBA-FIN-RE-1, *The Theft From Bangladesh Bank's Account* at the Federal Reserve Bank of New York, February 2016 (Section 5.5); DBA-FIN-RE-2, *The Outage of the Bank of England's RTGS System*, 20 October 2014 (Section 5.1); and DBA-FIN-RE-3, *Ukraine's Payment System Through the War*, 2022 to 2023 (Section 6.3). Their instruments were written by Project 11 on 14 October 2026, and this revision takes in their two errata and eight recommendations. Two events the plan for closing the seven sectors named for this sector are held for the classes they belong to, on the author's decision of the same date: the ransomware attack on the United States arm of the Industrial and Commercial Bank of China in November 2023, which disrupted the settlement of Treasury-market trades, for *Market Infrastructure*; and the denial-of-service attacks on the online services of United States banks in 2012 and 2013, for *Core Banking and Ledger*. The facility-class use case for the governing military-action scenario of Section 6.3 is DBA-FIN-UC-1, issued under this revision in the fifth wave of the plan.

10. Gaps and Open Items

10.1 Quantitative Design Parameters

The five parameters of Section 3.3 are stated without values. Their values for a system are to be validated against the operator's settlement records, its incident and exercise reports and its contingency rehearsals, and for a participant against its payment records and the deadlines its critical payments serve.

10.2 The Settlement-Finality Window

The framework proposes the settlement-finality window as the sector's consequence measure and leaves its calibration and escalation to this class (framework §3.6). Section 3.3 calibrates it as five parameters, the finality window, recovery time, integrity verification time, extension capacity and stranded value, with three zones of escalation. Whether the framework adopts the five and the zones for the sector, or keeps the window as the single measure and treats the others as this class's, is the framework's to decide, and the market-infrastructure class, whose securities settlement cycle is the window's other bound (framework §3.6), should be read against them before it does.

10.3 Retail Payment Systems

On the author's decision of 14 October 2026, retail payment systems are stated at the depth of the sub-index. A later revision is to develop them: the automated clearing house and the payroll it carries; the card networks and the public-facing consequence of a failure of authorization; and the instant payment systems that settle at all hours, which carry the first tier's requirements where they settle in central-bank money. A card-network outage, in which purchases could not be authorized across a region for some hours, is the kind of event that revision should read.

10.4 The Home of the Messaging Network

On the author's decision of 14 October 2026, the participant's endpoint is this class's and the financial messaging network is stated as a boundary. The network carries instructions between participants and to correspondents and is operated as a cooperative service with its own security programme; it is not a settlement system, and it is not a communications network in the sense of the communications series. Whether it belongs here, to the communications series or to the information-technology series is open. Its exclusion of participants by government decision is a military-action concern of this class whatever its home (Section 6.4).

10.5 What One Action Is for a Payment

The question this class joined at v0.1 is answered for the release case and not for the payment (Section 7.3). On the author's determination of 14 October 2026 on vendor releases and staged releases (*Three Author Decisions: Continuity Classes, Vendor Releases and Staged Releases*), *The Bright Line* §4.3 is not extended to a vendor's release, and the decision does not reach a

payment instruction, which is a participant's act at its own endpoint. The gate before finality rests on the framework's placement at §5, which stands. Whether §4.3 should be widened for the logical classes is not decided, and the class does not put the question now.

10.6 Resumption and Integrity

Section 3.3 states the tension between the two-hour aim and the time needed to verify records, and Sections 4.6 and 5.2 require it to be stated and decided in advance. The international guidance recognizes the tension without resolving it. Whether the sector framework should state a rule, such as that a first-tier system resumes on verified records unless a named authority decides otherwise on a stated basis, is for the framework. The guidance has since moved toward it: the CPMI-IOSCO toolkit of September 2026 treats resumption within two hours as likely conditional on the integrity of data, and Euroclear's response of 23 February 2016 questioned whether an uncorrupted copy of critical data was realistic for real-time systems, as *The Outage of the Bank of England's RTGS System* records both.

10.7 The Reference Events

The three reference events this class proposed are conferred by the framework at §9.1 as DBA-FIN-RE-1 to DBA-FIN-RE-3, and their instruments were written by Project 11 on 14 October 2026 and moved to v0.2 for DBA-FIN-RE-1 and RE-3 on the author's rulings of that date. DBA-FIN-RE-1 records Bangladesh Bank as military action without the military-action tier, a state's covert theft operation on the United States government's attribution, with denial not present; DBA-FIN-RE-2 records the RTGS outage as non-adversarial; and DBA-FIN-RE-3 records Ukraine's payment system as meeting the military-action threshold, all four characteristics established on the record, with Energy an impacted sector, and the record open. The instruments name institutions and not individuals. This revision takes in their errata at Sections 4.4, 4.5, 5.1, 5.5 and 6.3 and their recommendations at Sections 3.3, 4.3, 4.4, 4.5, 5.1, 5.5 and 6.3. The National Bank of Ukraine's annual report for 2022 could not be retrieved, and DBA-FIN-RE-3 records it as open. The use case for the governing scenario of Section 6.3 is DBA-FIN-UC-1.

10.8 The Register Row

The DBA Facility Class Register now records this class at v0.1 as an issued working draft under the framework at v0.4; Project 0 entered the row on 14 October 2026.

10.9 Errata and Notes Carried

The class took in at this revision the two errata Project 11 returned for it on 14 October 2026, with its Financial Services batch, at Sections 4.4, 4.5, 5.1, 5.5 and 6.3. None is outstanding. Its own notes to other instruments are at Section 10.10.

10.10 What This Class Notes to Other Instruments

Several elements of this instrument belong to other instruments, and this instrument records them as noted without stating them for those instruments. To the framework: the five parameters and three zones of the settlement-finality window (Section 3.3, Section 10.2), the event categories of Section 5, the three tiers of the sub-index with the participant's endpoint carried with its system (Section 2.2), the restated geolocation factor of Section 6.1, the declared residuals of Section 6.8, the conditionality principle of Section 6.9, and the resumption rule of Section 10.6; the conferral section it also asked for is now framework §9.1. To *Market Infrastructure* and *Core Banking and Ledger*, when written: the events held for them (Section 9.2) and the boundaries of Section 8.3; and to *Core Banking and Ledger*, from the instruments, *The Theft From Bangladesh Bank's Account* R-1, that a stop reaches every party at every hour, and *Ukraine's Payment System Through the War* R-1, that the Assured Finality Condition reaches the edge, which the retail revision of Section 10.3 also carries. To the communications series: that a first-tier settlement system's requirement for time during denial of the timing service is the duration of its longest settlement cycle (Section 5.4), for *Positioning, Navigation and Timing* to read beside its holdover. To the information-technology series: that a settlement system's software is a first-tier product and a second-tier participant's endpoint software a second-tier product in the sense of *Software, Hardware and Product Supply Chain* §2.2. To the data-center series: that the reversed elevation reaches the substrate of a second-tier participant's endpoint (Section 8.2). The register row is entered (Section 10.8) and the joined question is answered for the release case (Section 10.5).

References

DBA-FIN Series and Parent Framework

Financial Services Sector Framework (DBA-FIN_Sector_Framework). Eclectic Technologies, 2026. Parent sector framework under which this instrument is issued.

DBA-MA Parent Framework. Eclectic Technologies. Source of the design-basis-under-military-action methodology. *Design Basis Layers* (FD-DBL), held for the edition following the October 2026 Edition, is the canonical statement of the four military-action characteristics (§4.4, §6.4).

Defense in Depth (DBA-MA-DiD). Eclectic Technologies, 2026. Declared-residual output form (§3.5).

Software, Hardware and Product Supply Chain (DBA-IT-SW-FC3) and *Positioning, Navigation and Timing* (DBA-COMM-PN-FC4). Eclectic Technologies, 2026. The first logical class, whose structure this instrument follows, and the timing service it inherits.

The Theft From Bangladesh Bank's Account (DBA-FIN-RE-1), *The Outage of the Bank of England's RTGS System* (DBA-FIN-RE-2) and *Ukraine's Payment System Through the War*

(DBA-FIN-RE-3). Eclectic Technologies, 2026. Held by Project 11; conferred at Financial Services Sector Framework §9.1 and cited by short title and finding or recommendation number.

Facility-Class Types Are Physical, Coordination and Logical, author's determination of 30 September 2026, and *A Protective Function Running Logic Other Than What Was Tested Is Not Covered by the Carve-Out*, author's determination of 8 October 2026, both recorded by Project 0. *Closing the Seven Sectors*, working plan of 13 October 2026, Project 9. *Three Author Decisions: Continuity Classes, Vendor Releases and Staged Releases*, author's determination of 14 October 2026, recorded by Project 0.

Foundational Definitions and Command Broker Family

FD-BL, FD-BR, FD-LD, FD-LS, FD-EV, FD-BL-D1 and FD-BL-D2, cited by designation; version state in FD-REG (*Governing Stack Register*). CB-Framework_Cross_Sector (*Command Broker Framework*), cited by designation; version state in CB-REG (*Command Broker Family Register*).

Regulatory, Standards and Guidance

Committee on Payment and Settlement Systems and Technical Committee of the International Organization of Securities Commissions. *Principles for Financial Market Infrastructures*, April 2012, Principles 8 and 17. Committee on Payments and Market Infrastructures and International Organization of Securities Commissions. *Guidance on Cyber Resilience for Financial Market Infrastructures*, June 2016. CLS Bank International. Response to the consultative report on the guidance, 23 February 2016. Committee on Payments and Market Infrastructures and International Organization of Securities Commissions. *Cyber resilience toolkit: practical considerations for FMIs*, September 2026, as recorded in DBA-FIN-RE-2. Euroclear SA/NV. Response to the consultative report on the guidance, 23 February 2016.

Dodd-Frank Wall Street Reform and Consumer Protection Act, Title VIII, *Payment, Clearing, and Settlement Supervision Act of 2010*. Financial Stability Oversight Council. Designation of eight financial market utilities, 18 July 2012. Board of Governors of the Federal Reserve System. Regulation HH, 12 CFR Part 234; Regulation J, 12 CFR Part 210, and the Fedwire Funds Service operating circular.

Directive 98/26/EC of the European Parliament and of the Council on settlement finality in payment and securities settlement systems. Regulation (EU) 2022/2554 of the European Parliament and of the Council on digital operational resilience for the financial sector (DORA). Council Regulation (EU) 2022/345 of 1 March 2022, concerning restrictive measures in view of Russia's actions destabilising the situation in Ukraine. The messaging network's Customer Security Programme and Customer Security Controls Framework, 2016 onward.

Events and Investigations

Deloitte LLP. *Independent Review of RTGS Outage on 20 October 2014*, for the Court of the Bank of England, March 2015, and the Bank of England's response.

United States Department of Justice. Press release on the criminal complaint concerning cyber attacks attributed to a group linked to the government of North Korea, including the theft from Bangladesh Bank, 6 September 2018. Cybersecurity and Infrastructure Security Agency and partners. Alert AA20-239A, *FASTCash 2.0: North Korea's BeagleBoyz Robbing Banks*, 26 August 2020. Contemporary reporting on the theft, as cited in the text.

European Payments Council. *Payments in wartime: the story of the National Bank of Ukraine*, interview with the National Bank of Ukraine's Director of Payment Systems and Innovation Development, 28 July 2023. National Bank of Ukraine, statements on the operation of the System of Electronic Payments.

Revision History

Version	Date	Author	Description
0.1 — DRAFT	October 2026	Tim Roxey	Initial issue. First facility-class instrument in the DBA-FIN series and the second logical class written in the corpus, typed logical under the author's determination of 30 September 2026, in the ten-section structure of DBA-IT-SW-FC3, DBA-HPH-CD-FC1 and DBA-CHEM-PR-FC1. Written under the plan Closing the Seven Sectors, wave 2, on the author's decisions of 14 October 2026: large-value and wholesale settlement and the participant's interface in full, retail at sub-index depth; the participant's endpoint in this class and the messaging network a boundary left open; three reference events proposed, Bangladesh Bank (2016), the Bank of England RTGS outage (2014) and Ukraine's payment system through the war (from 2022), with ICBC (2023) held for Market Infrastructure and the 2012-13 denial-of-service attacks held for Core Banking. Calibrates the settlement-finality window the framework proposes as five parameters and three zones, states the instruction and the moment of finality as the governing pathway, and the Assured Finality Condition. Checked before issue by an independent review agent, not a person, against primary sources and for internal consistency; its fourteen findings were taken in, among them the review's own reasons for the 2014 contingency decision, the classification of the event categories, and the release gate restated by consequence rather than by value.
0.2 — DRAFT	October 2026	Tim Roxey	Conforms to the framework at v0.5 and takes in the return of Project 11's Financial Services batch, the author's rulings of 14 October 2026 on state theft and on reading the characteristics against a class's dependencies, and the author's determination of the same date on vendor releases. The three reference events

Version	Date	Author	Description
			proposed at v0.1 are stated as conferred, as DBA-FIN-RE-1 (Bangladesh Bank), DBA-FIN-RE-2 (the RTGS outage) and DBA-FIN-RE-3 (Ukraine), at Sections 1.3, 5.1, 5.5, 6.3, 9.2 and 10.7. Erratum 1 carries the rest of the review's paragraph 81 and its paragraph 60 finding, with the reasons of paragraphs 60 and 78 (Sections 4.5, 5.1). Erratum 2 gives the share of interbank payments by source and date and the branch count as of July 2023 (Section 6.3). The optional refinements read the thirty unpaid instructions as an outcome and time the query (Sections 4.4, 5.5). The eight recommendations enter at Sections 3.3 (the verification prepared), 4.3 (a failure of the endpoint's reporting as a sign), 4.4 (the receiving system's check before finality; a stop at every hour), 4.5 (the deadlines before the decider; the hour the decision to keep fixing expires) and 6.3 (the condition reaching the edge; permissions granted before the war). Sections 6.2, 7.3 and 10.5 record the rulings and the determination; the gate before finality stands on the framework's placement. Sections 10.8 to 10.10 record the register row, the errata taken in and the notes to Core Banking and Ledger. The cover lineage stays as issued. No change to the class's scope, chains, parameters, zones, event categories or residuals; the v0.1 entry is left as a record.